APPENDIX A - CORE PRACTICE    TIERED SCAFFOLDING    SOLUTIONS IN APPENDIX B

# {{ CHAPTER_TITLE: Comprehensive Topic Mastery & Practice Items }}

**TIER 1: GUIDED PRACTICE**    **ITEM: {{ ITEM_ID: PRAC-01 }}**

**{{ CAPABILITY_REF: CAP-ASSIGN-OXIDATION-NUMBER-CORE }}**

{{ PROMPT }}: Determine the oxidation state of Phosphorus in H3PO4.
Representation Spec: FORMULA_ANATOMY_VIEW

**Step-by-Step Guided Scaffolding:**

1. Identify fixed-rule elements: Hydrogen is in non-metal compound => H = +1.
   Three Hydrogen atoms contribute: 3 x (+1) = +3.
2. Identify Oxygen default: No peroxide bond present => O = -2.
   Four Oxygen atoms contribute: 4 x (-2) = -8.
3. Set up the Sum Invariant (Net charge = 0):
   (+3) + P + (-8) = 0 => P - 5 = 0 => P = [ ____ ].

**Solution Reference: [See Appendix B --> SOL-PRAC-01]**

**TIER 2: FADED PRACTICE**    **ITEM: {{ ITEM_ID: PRAC-02 }}**

**{{ CAPABILITY_REF: CAP-TRACK-ELEMENT-CHARGE-CHANGE }}**

{{ PROMPT }}: For the reaction: Zn(s) + 2HCl(aq) --> ZnCl2(aq) + H2(g),
identify which element undergoes oxidation and which undergoes reduction.

**Faded Support Steps:**

1. Write ON above each atom in reactants: Zn = [ ___ ], H in HCl = [ ___ ], Cl in HCl = [ ___ ].
2. Write ON above each atom in products: Zn in ZnCl2 = [ ___ ], Cl = [ ___ ], H in H2 = [ ___ ].
3. Track direction of change:
   * Zn changes from ___ to ___ => (Increase/Decrease) => [ _______________ ].
   * H changes from ___ to ___ => (Increase/Decrease) => [ _______________ ].

**Solution Reference: [See Appendix B --> SOL-PRAC-02]**

**TIER 3: INDEPENDENT PRACTICE**    **ITEM: {{ ITEM_ID: PRAC-03 }}**

**{{ CAPABILITY_REF: CAP-ATTACH-REDOX-ROLES }}**

{{ PROMPT }}: In the industrial reaction: Fe2O3(s) + 3CO(g) --> 2Fe(s) + 3CO2(g):
(a) Name the oxidising agent and reducing agent.
(b) Calculate the total moles of electrons transferred per mole of Fe2O3 reacted.

**Instructions: Complete without hints. Show your work clearly.**

Your Working Area:
* Oxidising Agent: ________________    Reducing Agent: ________________
* Electron Transfer Calculation: ________________________________________

**Solution Reference: [See Appendix B --> SOL-PRAC-03]**

**TIER 4: DIAGNOSTIC PROBE (TRAP DETECTOR)**    **ITEM: {{ ITEM_ID: PRAC-04 }}**

**{{ CAPABILITY_REF: CAP-DISCRIMINATE-PEROXIDE-EXCEPTION }}**

{{ PROMPT }}: A student calculates the oxidation state of Oxygen in BaO2 by applying the standard rule (O = -2), concluding Ba = +4. Evaluate this claim.

**Diagnostic Evaluation Questions:**

1. Is Ba = +4 physically permissible for an Alkaline Earth metal (Group 2)? [ YES / NO ]
2. Which rule takes priority: Group 2 fixed charge (+2) OR default Oxygen (-2)?
   Priority Rule: ________________________________________________________
3. Deduce the correct oxidation state of Oxygen in barium peroxide (BaO2):
   Correct ON of Oxygen = [ ____ ].

**Solution Reference: [See Appendix B --> SOL-PRAC-04]**